Input Encoders for Multi-Channel Signal Transformers: An Empirical Audit of Subspace Geometry

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Abstract

Transformers consuming multi-channel scalar signals must embed C simultaneous values into one d_{model}-dimensional vector per time step. We compare eight input encoders—spanning per-channel linear projections, an orthogonality regulariser, a nonlinear MLP stem, block-partitioned concatenation, channel-independent and channel-as-token architectures, and a projected positional encoding—on a synthetic benchmark where channel identity is informative and on ETTh1 as a real-data check, with most comparisons at 20 paired seeds. At convergence the standard per-channel linear projection (`nn.Linear(C,` d_{model}`)`) and its neighbours (orthogonality-regularised, block-partitioned, MLP-stem) sit in a 0.02 NLL band of each other on a $\sim$2.16 NLL baseline at C=4. Paired-difference tests resolve several small inter-encoder gaps as statistically clear: LINEAR-PPE leads by Δ=0.041 at C=4 (paired p=2.6×10^{-6}, 19/20) and the lead shrinks but persists to C=16 (Δ=0.016, p=0.036, 14/20); a direct geometric probe shows LINEAR-PPE's mechanism is positional–channel orthogonalisation (fraction of positional energy inside the channel subspace drops $\sim$6.3×, 3.4% → 0.5%). MLP pulls ahead of the linear family at C=16 (paired $p \in [2\times10^{-4}, 5\times10^{-12}]$), with the gap shrinking $\sim$3× at 10× training data but staying clear. On ETTh1 CAT posts the lowest mean NLL, decisively above MLP and LINEAR-PPE but statistically tied with LINEAR and SUM-ORTHO. The picture is one of practical near-equivalence within the top tier: statistically real gaps that are small in magnitude and would not, on this evidence, be expected to translate cleanly to new tasks. The practical recommendation is to use `nn.Linear(C,` d_{model}`)`, optionally with the learned positional projection, and reach for something more elaborate only when there is a real and strong reason in the task at hand. Code: https://github.com/OssiLehtinen/FDM-encoding.

1 Introduction

Transformers applied to multivariate numerical signals—multi-sensor telemetry, financial factor stacks, biomedical waveforms—face a design choice at the input layer: how to embed C simultaneous scalar channels into one d_{model}-dimensional vector per time step.

Prior art. The additive embedding recipe inherited from NLP (1) projects each input to a vector and sums. When the inputs are multiple channels rather than one token-plus-position pair, this comes in two forms: a *shared* scalar projection W with per-channel additive embeddings, or a *per-channel* projection W_k—the latter being mathematically a single `nn.Linear(C,` d_{model}`)`. Many multivariate time-series transformers use a linear input stem (2, 3), though details vary. Alternative

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architectures treat each channel as a separate token (iTransformer (5)), run independent backbones per channel (PatchTST (4)), or combine both (Crossformer (6)). Orthogonality penalties on weight matrices have been used for feature decorrelation in deep networks (7).

Open questions. Despite the variety of approaches, several basic questions remain unanswered in the literature: (i) Does a shared-scalar projection with a small channel embedding actually work, or does it have a hard ceiling? (ii) Is block partitioning, an orthogonality penalty, or nonlinearity needed, or does a bare linear layer suffice? (iii) Does the task loss itself produce channel separation without explicit enforcement? (iv) Does the positional encoding interact with the channel encoding, and can this interaction be improved?

Our contribution. We compare eight encoders on a controlled synthetic benchmark designed to make channel identity informative, with supporting experiments on the public ETTh1 dataset (2). We find that: (i) the per-channel linear projection matches every alternative at convergence; (ii) a shared-scalar baseline has a sharp, capacity-independent ceiling; (iii) the task loss drives the learned W_k to near-orthogonal subspaces and scales their norms larger for outcome-relevant channels; and (iv) within the per-channel-W_k family the encoder landscape is practically near-equivalent: 20-seed paired tests resolve several small inter-encoder gaps as statistically clear, but the absolute differences are small relative to the spread typical task-design choices would induce. Three caveats: LINEAR-PPE gains $\sim$0.04 NLL over LINEAR at C=4 (paired p=2.6$\times$10^{-6}) and a smaller but still significant amount at C=16, with the mechanism (positional–channel orthogonalisation) directly measured; MLP pulls ahead at C=16 (paired $p \in [2\times10^{-4}, 5\times10^{-12}]$, gap shrinks 3$\times$ at 10$\times$ data); and CAT posts the lowest mean NLL on ETTh1 but is statistically tied with LINEAR and SUM-ORTHO.

2 Method

2.1 Model architecture

All encoders share a common backbone: a small causal transformer with d_{model}=64, 4 attention heads, 3 layers, feed-forward width d_{ff}=256, dropout 0.1, pre-LayerNorm, and a linear categorical head mapping to K=32 logits. The *only* component that varies across experiments is the input encoder (Figure 1).

2.2 Encoders

We study eight encoders. In the definitions below, $v_k(t)$ is the scalar value of channel k at time t, $\mathbf{p}(t)$ is a fixed sinusoidal positional encoding (1), and all learned parameters are randomly initialised.

sum (ablation baseline). Shared scalar projection $W \in \mathbb{R}^{d_{\text{model}}}$, learned per-channel additive embedding e_k:

$$\mathbf{h}(t) = \sum_k \big(W\, v_k(t) + e_k\big) + \mathbf{p}(t).$$

Total encoder capacity is $\mathcal{O}(d_{\text{model}})$ regardless of C. We use this to isolate what goes wrong when the input projection shares weights across channels.

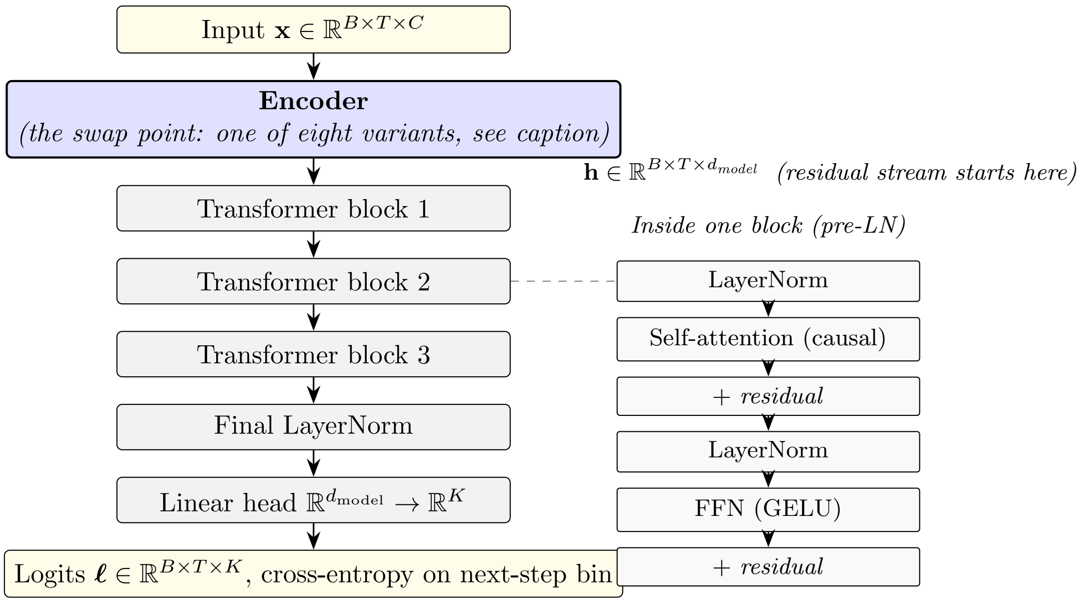

Figure 1: Architecture. Every experiment uses the same causal transformer backbone (three pre-LN blocks, each structured as in the inset on the right: LN → self-attention → residual, LN → FFN → residual; followed by a final LayerNorm and a linear head into K=32 bin logits). **The only component that varies across experiments is the encoder (blue).** Eight encoders share the same input/output signature $\mathbb{R}^{B\times T\times C} \to \mathbb{R}^{B\times T\times d_{\text{model}}}$ and are therefore drop-in interchangeable. The variants are: SUM (shared W then sum across channels), LINEAR (per-channel W_k then sum), SUM-ORTHO (LINEAR plus an orthogonality penalty on the W_k), MLP (two-layer MLP on the channel vector), LINEAR-PPE (LINEAR with a learned linear projection of the positional encoding), CONCAT (per-channel block concatenated), CI (channel-independent backbone), and CAT (channel-as-token attention). Precise definitions are in Section 2.

linear. Per-channel projection W_k and bias b_k, summed:

$$\mathbf{h}(t) = \sum_k \big(W_k\, v_k(t) + b_k\big) + \mathbf{p}(t).$$

Equivalently $\mathbf{h}(t) = W\,\mathbf{v}(t) + \mathbf{b} + \mathbf{p}(t)$ with $W \in \mathbb{R}^{d_{\text{model}}\times C}$: a single `nn.Linear(C, d_model)`.

sum-ortho. Same as LINEAR with an auxiliary loss $L_{\text{ortho}} = \lambda \sum_{i\neq j}(W_i \cdot W_j)^2/2$ (λ=10^{-2}). This variant isolates the regulariser's contribution.

linear-ppe (projected positional encoding). Same channel encoding as LINEAR, but the sinusoidal position passes through a learned linear layer:

$$\mathbf{h}(t) = \sum_k \big(W_k\, v_k(t) + b_k\big) + W_{\text{pos}}\,\mathbf{p}(t) + \mathbf{b}_{\text{pos}},$$

with $W_{\text{pos}} \in \mathbb{R}^{d_{\text{model}}\times d_{\text{model}}}$. This lets the optimiser rotate the positional subspace to be orthogonal to the channel subspace. Adds $d_{\text{model}}^2 + d_{\text{model}}$ parameters (4,160 at d=64).

mlp. Two-layer MLP on the channel vector: $\mathbf{h}(t) = W_2\,\mathrm{GELU}(W_1\,\mathbf{v}(t) + \mathbf{b}_1) + \mathbf{b}_2 + \mathbf{p}(t)$, with $W_1 \in \mathbb{R}^{d_{\text{model}} \times C}$, $W_2 \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$. Tests whether nonlinearity helps ($\sim 9\times$ more encoder parameters than LINEAR).

concat. Per-channel projection into $d_{\text{block}}{=}d_{\text{model}}/C$ dims, concatenated (requires $C \mid d_{\text{model}}$): $\mathbf{h}(t) = [\mathbf{e}_0(t) \| \cdots \| \mathbf{e}_{C-1}(t)] + \mathbf{p}(t)$. Channel identity is an architectural invariant.

ci (channel-independent). PatchTST-spirit (4): shared-weight causal transformer per channel; final hidden states concatenated into the head. Compute: $\mathcal{O}(C\,B\,T^2)$.

cat (channel-as-token). iTransformer-spirit (5): each (t, k) pair is a token, sequence length $C{\cdot}T$, causal across time, bidirectional within a time step. Attention: $\mathcal{O}((CT)^2)$; skipped at $C{=}16$.

2.3 Training protocol

Identical across encoders and across both datasets unless noted.

Objective. At every position $t \in \{0, \ldots, T{-}1\}$ the model emits logits $\ell_t \in \mathbb{R}^K$ and we minimise the next-step categorical cross-entropy

$$\mathcal{L}_{\text{NLL}} = -\frac{1}{B(T{-}1)} \sum_{b=1}^{B} \sum_{t=0}^{T-2} \log \mathrm{softmax}(\ell_t^{(b)})_{\,y_{t+1}^{(b)}},$$

where $y_t^{(b)}$ is the bin index at position t in series b. The predicted distribution at position t is supervised only by the bin at position $t+1$, and causal masking inside the transformer prevents the model from seeing y_{t+1} at prediction time. The SUM-ORTHO encoder adds the auxiliary loss $\lambda \sum_{i \neq j} (W_i \cdot W_j)^2/2$ with $\lambda{=}10^{-2}$; all other encoders have $\mathcal{L} = \mathcal{L}_{\text{NLL}}$.

Optimiser. AdamW with peak learning rate $3{\times}10^{-4}$, weight decay 10^{-4}, and PyTorch defaults for momentum ($\beta_1{=}0.9$, $\beta_2{=}0.999$, $\epsilon{=}10^{-8}$). Gradient norm is clipped to 1.0 per step. Batch size is 32. No warm-up. Learning rate follows a cosine annealing schedule from peak to $3{\times}10^{-6}$ (1% of peak) over the full 300 epochs.

Train/val split. For the synthetic benchmark we generate $n_{\text{series}}{=}512$ length-$T{=}160$ series per seed and split with a seeded `torch.utils.data.random_split` into 461/51 ($\approx$ 90/10) train and val. For ETTh1 we follow the dataset's standard chronological partition, using the train block to fit per-variate standardisation statistics and the quantile bin edges for the target, and evaluating on the val block. Data loaders use `shuffle=True` for training, `shuffle=False` for validation, batch size 32 in both cases.

Validation cadence and model selection. Validation NLL and top-1 bin accuracy are evaluated every 20 epochs, plus at epoch 1 and epoch 300 (giving a trace of 17 points). We do not actually halt training early; instead, we report the best validation NLL observed along this trace, together with the accuracy at that same epoch ("effective early stopping"). This decouples our headline metric from slight differences in convergence epoch across encoders while still letting every run see the full 300-epoch schedule.

Seeds and reproducibility. Most configurations are run over 20 random seeds, indexed 0 through 19 (the first five are the canonical paper runs; the next fifteen are paired-seed extras from an open-ended round-robin sweep). Single-seed-by-design diagnostics (linear probing, test-time channel masking) and the bias ablation are kept at the canonical 5 seeds. Seed s sets `torch.manual_seed(s)` (controlling weight initialisation and dropout) and seeds the `torch.Generator` passed to `random_split` (controlling the train/val partition). On the synthetic benchmark, seed s also drives the `numpy` RNG that generates the input series themselves. *The same five seeds are reused for every encoder variant, every C, and every d_{model}.* This is a paired-seed design: at a fixed s, every encoder sees identical training data and the identical train/val partition, and only the model architecture (and its initial parameter draws from the shared PRNG state) differs. We *report* unpaired mean ± std intervals throughout for readability and to match the conventions of the surrounding literature, but we *test* with paired-difference statistics where the unpaired interval is loose enough that paired analysis changes the conclusion. Concretely: where the paired and unpaired test agree qualitatively (most rows in this paper), we report only the unpaired interval; where they disagree, or where the unpaired p-value is borderline, we report the paired result and call it out. Three such comparisons appear: the LINEAR-PPE gap at C=4 (Section 3.10), the MLP lead at C=16 (Section 3.2), and the CAT lead on ETTh1 (Section 3.11). We do not enable `torch.use_deterministic_algorithms(True)`, so CUDA matmul nondeterminism contributes some additional seed-level variation between full re-runs; the reported ± std intervals capture this. All headline numbers in this paper are means ± sample standard deviations over the five seeds, computed at each run's best validation NLL epoch.

2.4 Datasets

Synthetic. N=512 time series of length T=160 with C channels. Each channel is a sum of three sinusoids with random frequencies in $[0.005, 0.08]$ cycles/sample plus AR(1) noise, standardised. The outcome is a fixed non-linear function of the first four channels with lags 3 and 7:

$$y_t = \tanh\big(s_0(t-3)\, s_1(t)\big) + 0.6 \sin\big(1.3\, s_2(t-7)\big) + 0.4\mathbb{1}[s_3(t) > 0]\, s_0(t),$$

binned into K=32 quantile bins. Channels $k \geq 4$ are independent distractors. The task is designed so that an encoder that mixes channels at the input layer cannot recover the interaction structure.

ETTh1. Electricity Transformer Temperature (2), hourly, 7 variates. We standardise on the train split, bin the target variate OT into 32 quantile bins, and predict the next-step bin given all 7 variates over T=160. The model uses d_{model}=56 (divisible by C=7), 7 heads, d_{ff}=224; training is otherwise identical.

2.5 Diagnostic analyses

Beyond aggregate NLL, we use three diagnostics to inspect what the per-channel-W_k encoders actually learn. Each is computed after full 300-epoch training, on five seeds unless stated otherwise.

Gram-matrix analysis. LINEAR and SUM-ORTHO differ only in whether the soft-orthogonality penalty $L_{\text{ortho}} = \lambda \sum_{i \neq j} (W_i \cdot W_j)^2/2$ is added to the loss. If both reach the same NLL, the question is whether LINEAR's W_k are geometrically similar to SUM-ORTHO's—i.e. whether the task pressure alone already produces a near-orthogonal arrangement—or whether the two encoders arrive at the same loss through different geometries.

After training we extract the per-channel projection matrix $W \in \mathbb{R}^{C\times d_{\text{model}}}$ (rows W_k) from the encoder and compute the cosine matrix $G_{ij} = (W_i \cdot W_j)/(\|W_i\|\|W_j\|)$. Two scalar summaries:

$$\overline{|\cos|} = \frac{1}{C(C-1)}\sum_{i\neq j}|G_{ij}|, \qquad \max|\cos| = \max_{i\neq j}|G_{ij}|.$$

Both go to zero iff the W_k become mutually orthogonal. We report seed-averaged means.

Encoding-space allocation. Orthogonality says channels are distinguishable; it does not say whether the model dedicates more representational capacity to outcome-relevant channels than to distractors. We measure two quantities, both at the encoder output:

(i) Norm $\|W_k\|$ for each channel. A larger norm means channel k's contribution $W_k v_k(t) + b_k$ has larger magnitude in the embedding, i.e. occupies more of the residual stream.

(ii) Per-channel variance fraction. The encoder output is a sum of per-channel contributions $W_k v_k(t) + b_k$. Treating each contribution as a random vector indexed by (t, batch), the per-channel variance is

$$\sigma_k^2 = \sum_{d=1}^{d_{\text{model}}} \operatorname{Var}_{t,\text{batch}}\big[(W_k v_k(t) + b_k)_d\big],$$

and we report fractions $\sigma_k^2/\sum_j \sigma_j^2$ on a held-out batch. This combines the size of W_k with the empirical variability of channel k's input distribution.

For both metrics, channels are partitioned into drivers ($k \in \{0, 1, 2, 3\}$, which enter the outcome formula) and distractors ($k \geq 4$, independent of the outcome). Reporting them this way lets us check whether the model independently discovered the driver/distractor split.

Linear-probing analysis. The encoder lays down a representation; the question is whether downstream attention and FFN layers preserve enough of it that each channel remains linearly identifiable several blocks deep. We test this by training closed-form ridge probes from a frozen hidden state back to the raw input.

Procedure: (1) train the encoder–transformer end-to-end as usual; (2) freeze it; (3) generate a fresh probe dataset (`MultiSignalDataset` with a different seed, so the model has not been trained on it); (4) for each layer of interest, pass the probe dataset through and collect hidden states $H \in \mathbb{R}^{N\times d_{\text{model}}}$ alongside the corresponding raw inputs $X \in \mathbb{R}^{N\times C}$; (5) solve the ridge regression

$$W_{\text{probe}} = (H^\top H + \lambda I)^{-1} H^\top X, \qquad \lambda = 10^{-3},$$

in closed form on a probe-train split; (6) evaluate held-out R^2 per channel on a probe-validation split: $R_k^2 = 1 - \text{SS}_{\text{res},k}/\text{SS}_{\text{tot},k}$.

We probe two layers: layer 0 (the encoder output, i.e. the input to the first transformer block) and layer 3 (the output of the third and final transformer block, before the LayerNorm and the categorical head). $R^2 = 1$ at layer 0 confirms the encoder writes the raw channel into a linearly recoverable subspace; R^2 at layer 3 reveals how much of that recoverability survives the depth of the model. CI and CAT are excluded because their hidden states have a fundamentally different shape (per-channel stream and per-token respectively), so a single $H \to X$ probe does not apply.

Table 1: Main comparison at C=4, d_{model}=64, 300 epochs, cosine LR decay, mean $\pm$ std over 20 seeds (5 canonical + 15 paired-seed extras). Best val NLL (effective early stopping). Random baseline: NLL $= \ln 32 \approx 3.47$, acc $=0.031$. †For CI and CAT, "encoder params" counts only the per-channel input projection (plus per-variate embedding for CAT); the bulk of those architectures' parameters lives in the replacement backbone.

Encoder	Enc. params	Val NLL ↓	Val acc ↑
SUM	320	3.257 ± 0.011	0.091 ± 0.003
CI†	128	3.053 ± 0.013	0.136 ± 0.004
CAT†	384	2.348 ± 0.060	0.212 ± 0.011
CONCAT	128	2.170 ± 0.025	0.243 ± 0.006
LINEAR	512	2.155 ± 0.019	0.247 ± 0.006
SUM-ORTHO	512	2.155 ± 0.020	0.247 ± 0.006
MLP	4,480	2.177 ± 0.022	0.242 ± 0.007
LINEAR-PPE	4,672	2.114 ± 0.029	0.256 ± 0.009

3 Results

3.1 Main comparison

Table 1 reports all eight encoders at C=4, d_{model}=64, 300 epochs with cosine LR decay, 20 seeds. The headline read is *practical near-equivalence among the per-channel-W_k family*: LINEAR, SUM-ORTHO, CONCAT, and MLP all fall in a 0.02 NLL band at 2.155–2.177, with LINEAR and SUM-ORTHO matching to three decimals (the orthogonality penalty contributes nothing at convergence). LINEAR-PPE sits a touch below at 2.114, a ~0.04 NLL improvement over LINEAR that is small in magnitude but statistically solid under the paired-seed design: 19 of 20 paired differences favour LINEAR-PPE (paired t-test p=2.6×10^{-6}, 95% bootstrap CI on Δ NLL [+0.029, +0.053]). The mechanism behind the gap is direct: a learned linear projection of the positional encoding rotates it out of the channel subspace (Section 3.10). In practical terms the gap from LINEAR to LINEAR-PPE corresponds to about 2% of the baseline NLL and is well within the spread that small task-design choices would induce; we read it as a real effect, not a practically transformative one. CI and CAT underperform despite nominally giving each channel more capacity. The only encoder that loses decisively is SUM, which lacks per-channel projections.

3.2 Channel-count scaling

All encoders degrade as C grows (channels $k \geq 4$ are distractors). The per-channel-W_k family stays clustered at the top across $C \in \{4, 8, 16\}$ and the gap to SUM widens rather than narrows. LINEAR-PPE sits at the very top across all three C values, with paired tests at 20 seeds: $\Delta = 0.041$, p=2.6×10^{-6} at C=4 (19/20 favour ppe); $\Delta = 0.026$, p=3.7×10^{-4} at C=8 (16/20); $\Delta = 0.016$, p=0.036 at C=16 (14/20). The advantage shrinks with C but does not vanish; even at C=16 the paired test puts the gap above zero by a small margin.

At C=16, MLP pulls ahead of the linear family (2.231 vs. 2.27–2.34). Paired tests against LINEAR (p=2×10^{-4}, 18/20), SUM-ORTHO (p=7×10^{-4}, 17/20), and CONCAT (p=5×10^{-12}, 20/20) are decisive; against LINEAR-PPE the gap is small and the call is borderline ($\Delta = 0.020$, paired p=0.053, 12/20). Combined with the LINEAR-PPE lead at smaller C, the picture is a narrow top tier whose member ordering shifts with C: LINEAR-PPE edges the linear family at every C, MLP edges the linear family

Table 2: Scaling with channel count. Channels 0–3 drive the outcome; additional channels are independent distractors. 300 epochs with cosine decay, mean ± std over 20 seeds. CAT is skipped at C=16 because its $\mathcal{O}((CT)^2)$ attention cost explodes. Sub-band sub-tables span the within-cell standard error times a t-quantile; encoder-vs-encoder paired tests reported in the text use the same paired-seed design across encoders.

C	encoder	val NLL ↓	val acc ↑
4	SUM	3.257 ± 0.011	0.091 ± 0.003
	CI	3.053 ± 0.013	0.136 ± 0.004
	CAT	2.348 ± 0.060	0.212 ± 0.011
	CONCAT	2.170 ± 0.025	0.243 ± 0.006
	MLP	2.177 ± 0.022	0.242 ± 0.007
	SUM-ORTHO	2.155 ± 0.020	0.247 ± 0.006
	LINEAR	2.155 ± 0.019	0.247 ± 0.006
	LINEAR-PPE	2.114 ± 0.029	0.256 ± 0.009
8	SUM	3.300 ± 0.006	0.081 ± 0.004
	CI	3.062 ± 0.016	0.135 ± 0.004
	CAT	2.334 ± 0.073	0.213 ± 0.011
	CONCAT	2.234 ± 0.023	0.231 ± 0.005
	MLP	2.211 ± 0.027	0.236 ± 0.005
	SUM-ORTHO	2.176 ± 0.022	0.244 ± 0.006
	LINEAR	2.176 ± 0.022	0.243 ± 0.006
	LINEAR-PPE	2.150 ± 0.030	0.249 ± 0.006
16	SUM	3.310 ± 0.004	0.075 ± 0.003
	CI	3.078 ± 0.017	0.133 ± 0.005
	CONCAT	2.340 ± 0.033	0.215 ± 0.008
	SUM-ORTHO	2.267 ± 0.038	0.227 ± 0.007
	LINEAR	2.267 ± 0.032	0.226 ± 0.007
	LINEAR-PPE	2.252 ± 0.038	0.230 ± 0.008
	MLP	2.231 ± 0.032	0.234 ± 0.004

(but ties LINEAR-PPE) at C=16, and at every C the absolute spread among the top four is on the order of 0.02–0.05 NLL on a baseline of ∼2.2 NLL.

The C=16 ranking persists, but shrinks, with 10× training data. The ΔNLL ∼0.07 between MLP and LINEAR at C=16, N_{series}=512 raises an obvious question: is the nonlinear stem genuinely useful, or is the main sweep data-limited and MLP just happens to extract more from sparse data? We test this directly with the `main_largen` stage of our reproducer: the same top-tier encoders at C=16 trained with N_{series}=5120 (10× the main-sweep data), 20 seeds, otherwise identical.

Two things happen at once. (i) MLP's lead survives in the data-rich regime: 18–20 of 20 paired-seed differences favour MLP over each of LINEAR (paired p=4×10^{-4}), SUM-ORTHO (p=9.6×10^{-4}), LINEAR-PPE (p=2.4×10^{-4}), and CONCAT (p=2.3×10^{-13}). The C=16 MLP finding is not a data-limited artefact. (ii) The magnitude shrinks roughly three-fold: ΔNLL(MLP–LINEAR) goes from 0.036 at N=512 to 0.011 at N=5120; the full linear family at N=5120 falls inside a band of width 0.024 NLL versus 0.073 at N=512. Each encoder also gains ∼0.4 NLL from the extra data, confirming the N=512 sweep is data-limited at C=16 just as at C=4 (Section 3.6). The encoder ordering at

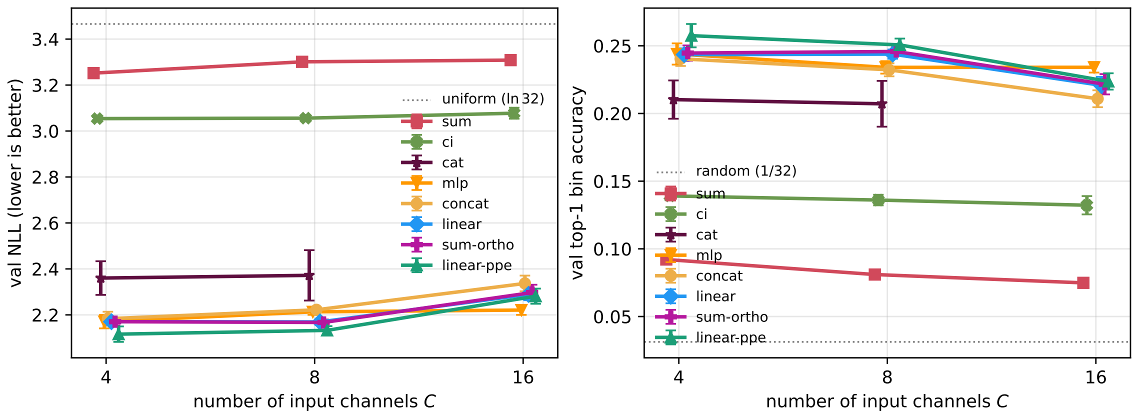

Figure 2: Scaling with input channel count C (error bars: one seed std over 20 seeds). CAT skipped at C=16 due to $\mathcal{O}((CT)^2)$ attention cost.

Table 3: C=16 top-tier encoders, N_{series}=5120 vs. N_{series}=512 (20 seeds each, 300 epochs). The full linear family at N=5120 falls inside a 0.024 NLL band.

encoder	N=512 val NLL	N=5120 val NLL	Δ from data
MLP	2.231 ± 0.032	1.831 ± 0.009	-0.400
LINEAR-PPE	2.252 ± 0.038	1.841 ± 0.013	-0.411
LINEAR	2.267 ± 0.032	1.843 ± 0.015	-0.424
SUM-ORTHO	2.267 ± 0.038	1.844 ± 0.018	-0.423
CONCAT	2.340 ± 0.033	1.854 ± 0.011	-0.486

C=16 is robust to data scale; the absolute size of the gaps is not.

3.3 Model-width scaling

Table 4 varies d_{model} over $\{64, 128, 256\}$ at C=4, 20 seeds, all top-tier encoders plus the SUM and CONCAT baselines.

Three findings. First, SUM's ceiling is structural and capacity-independent: 15× more parameters move NLL by less than 0.01, as expected from the information-theoretic argument later in Section 3.5.

Second, *every* encoder shows clear overfitting at $d_{\text{model}} \geq 128$. Best val NLL is reached at epoch ∼100 (d=128) or ∼40 (d=256), after which val NLL drifts upward (catastrophically for MLP at d=256, where the end-of-training val NLL exceeds $\ln 32$). Effective early stopping (Section 2.3) is what protects the reported best-NLL numbers; the trajectories themselves are not converged. C=4 with N_{series}=512 does not justify a 256-dim residual stream.

Third, the linear family's best-val ceiling rises gently with d_{model} ($2.16 \to 2.23 \to 2.32$ for LINEAR), indicating that the optimal effective-stopping model gets worse as the model is given more capacity to overfit. MLP degrades fastest of the top tier ($2.18 \to 2.30 \to 2.41$) because its encoder parameters scale as d^2_{model} and exacerbate the capacity mismatch. The paired tests against LINEAR sharpen with width: p=2×10^{-4} at d=64 (18/20 favour LINEAR), p=4×10^{-9} at d=128 (20/20), and p=5×10^{-9} at d=256 (19/20). The C=4 small-channel regime does not reward nonlinearity, and gives it room to overfit when widened. This is in sharp contrast to C=16 where MLP leads

Table 4: Scaling with d_{model} at C=4. d_{ff}=4 d_{model}, 4 attention heads (so d_{head} grows). Mean ± std over 20 seeds, 300 epochs with cosine decay. *All* encoders flag as overfitting at $d_{\text{model}} \geq 128$: best val NLL is reached well before epoch 300 (typical best epochs ∼100 at d=128, ∼40 at d=256), and val NLL drifts upward afterwards. The reported best-NLL numbers are the effective-early-stopping minima.

d_{model}	encoder	total params	val NLL ↓	val acc ↑
64	SUM	152,480	3.257 ± 0.011	0.091 ± 0.003
	CONCAT	152,288	2.170 ± 0.026	0.243 ± 0.006
	LINEAR	152,672	2.155 ± 0.019	0.248 ± 0.005
	SUM-ORTHO	152,672	2.155 ± 0.019	0.248 ± 0.005
	MLP	156,640	2.177 ± 0.023	0.241 ± 0.008
	LINEAR-PPE	156,832	2.115 ± 0.029	0.256 ± 0.009
128	SUM	599,840	3.260 ± 0.010	0.091 ± 0.003
	CONCAT	599,456	2.229 ± 0.031	0.235 ± 0.006
	LINEAR	600,224	2.234 ± 0.032	0.235 ± 0.006
	SUM-ORTHO	600,224	2.233 ± 0.032	0.235 ± 0.007
	MLP	616,352	2.301 ± 0.027	0.222 ± 0.007
	LINEAR-PPE	616,736	2.185 ± 0.032	0.245 ± 0.005
256	SUM	2,379,296	3.262 ± 0.011	0.091 ± 0.003
	CONCAT	2,378,528	2.311 ± 0.026	0.217 ± 0.004
	LINEAR	2,380,064	2.322 ± 0.033	0.217 ± 0.006
	SUM-ORTHO	2,380,064	2.323 ± 0.036	0.217 ± 0.006
	MLP	2,445,088	2.407 ± 0.023	0.210 ± 0.007
	LINEAR-PPE	2,445,856	2.265 ± 0.026	0.229 ± 0.006

(Section 3.2): the high-channel-pressure regime is what makes nonlinearity useful.

linear-ppe continues to lead. At every d_{model} tested, LINEAR-PPE achieves the lowest val NLL. The paired gaps to LINEAR are Δ=0.040 at d=64 (paired p=3×10^{-6}, 19/20), Δ=0.049 at d=128 (paired p=3×10^{-4}, 18/20), and Δ=0.056 at d=256 (paired p=4×10^{-8}, 19/20). The advantage is robust (56/60 paired-seed comparisons across the three widths favour LINEAR-PPE), and the magnitude grows mildly with d_{model}. Both observations are consistent with the direct geometric measurement in Section 3.10: the rotation is doing real work, and more residual-stream capacity gives it more directions to rotate into.

3.4 Gram-matrix analysis: spontaneous orthogonality

LINEAR has no mechanism pushing the W_k toward orthogonality; SUM-ORTHO adds an explicit penalty. Both produce the same NLL to three decimals. Applying the off-diagonal cosine diagnostic from Section 2.5 gives:

Without any penalty the learned W_k are already near-orthogonal (mean off-diagonal cosine 0.042, 95% bootstrap CI $[0.037, 0.046]$). The regulariser tightens this by another factor of ∼4 to 0.010 (95% CI $[0.008, 0.011]$) but does not improve downstream NLL. The task loss itself provides enough gradient pressure to separate the per-channel projections once they exist as free parameters.

Table 5: Learned per-channel W_k geometry (C=4, 300 epochs, mean over 20 seeds).

encoder	best val NLL	$\max_{i \neq j} \lvert \cos(W_i, W_j) \rvert$	mean $\lvert \cos \rvert$
LINEAR	2.155	0.095	0.042
SUM-ORTHO	2.155	0.022	0.010

3.5 Bias ablation: the biases play no role in the geometry

LINEAR's forward pass is $h(t) = \sum_k (W_k v_k(t) + b_k) + \mathbf{p}(t)$. Distributing the sum, the per-channel biases collapse into a single d_{model}-dimensional constant offset $B = \sum_k b_k$ that is added to every embedding regardless of input or position. Algebraically, the $C \times d_{\text{model}}$ bias parameters are an over-parameterised reparameterisation of one global offset, and they never enter the cosine computation in Table 5. To confirm experimentally that they play no role in the orthogonalisation, we retrain LINEAR with the `channel_bias` parameter zeroed and frozen (LINEAR-NOBIAS):

Table 6: Channel-bias ablation (C=4, 300 epochs, 5 seeds).

encoder	val NLL ↓	mean $\lvert \cos \rvert$	max $\lvert \cos \rvert$
LINEAR	2.169 ± 0.014	0.036	0.082
LINEAR-NOBIAS	2.177 ± 0.013	0.036	0.083

NLL is unchanged within seed std, and the Gram statistics match to three decimals. The $C \cdot d_{\text{model}} = 256$ bias parameters at this configuration are 0.17% of the full model's ∼152K parameters, so the saving is too small to drive a practical recommendation; we retain the bias in the headline definition of LINEAR to match the PyTorch `nn.Linear` default. The point of this ablation is epistemic: it confirms that the geometric argument in this paper is bias-independent.

3.6 Encoding space allocation

Beyond separating channels, does the model allocate more encoding space to channels that matter more? Using the norm and variance-fraction diagnostics from Section 2.5, at $C \in \{4, 8, 16\}$:

Table 7: Encoding space allocation for LINEAR (300 epochs, mean over 20 seeds). Channels 0–3 drive the outcome; the rest are distractors.

	$\lVert W_k \rVert$ norm		variance fraction	
	drivers (0–3)	distractors	drivers	distractors
C=8	1.22	0.54	83.8%	16.2%
C=16	1.30	0.59	62.4%	37.6%

Driver channels receive ∼2.2× larger seed-averaged norms than distractors at C=8 and ∼2.1× at C=16. The four drivers capture 83% of the embedding variance at C=8 and 61% at C=16, where the aggregate share is naturally diluted by having twelve distractors but the per-channel variance contribution remains markedly larger for drivers (∼15% each vs. ∼3% each). We do not report per-channel error bars on the norms, and the fine-grained ordering within drivers is descriptive only; the robust claim is the driver/distractor gap, which holds at both channel counts.

Why distractor norms are not zero. The driver/distractor gap is large but the distractor norms ($\sim$0.55 at C=8) do not vanish. This is consistent with the finite-data dynamics rather than an asymptotic property of the encoder. The expected gradient on a distractor weight W_k is

$$\mathbb{E}\left[\frac{\partial \mathcal{L}}{\partial W_k}\right] = \mathbb{E}\left[\frac{\partial \mathcal{L}}{\partial \mathbf{h}(t)} \cdot v_k(t)\right],$$

which factorises to zero whenever the distractor input v_k is independent of the drivers and the outcome (true by construction in the synthetic benchmark, and approximately so for the standardised synthetic channels). With finite data, however, stochastic gradient estimates have nonzero variance whose magnitude scales as $\mathcal{O}(1/\sqrt{N_{\text{series}}})$. Under L2 weight decay (coefficient λ=10^{-4}), distractor weights reach an equilibrium whose magnitude is approximately $\sqrt{\sigma^2_{\text{noise}}/\lambda}$, i.e. they sit at a noise floor rather than being driven to zero.

Several interventions would push distractor norms down further. (i) More training data: $1/\sqrt{N}$ scaling predicts that 10$\times$ the data should shrink distractor norms by $\sim\sqrt{10} \approx 3.2$, from $\sim$0.55 toward $\sim$0.17, with driver norms roughly unchanged. Running `linear` at C=8 with N_{series}=5120 (single seed; the `geom_largen` stage of our reproducer, output in `results_paper/geom_largen.json`) confirms this qualitatively: the distractor mean norm drops to 0.218 (a $\sim$2.5$\times$ reduction, close to but slightly slower than the predicted $\sim$3.2), the driver/distractor norm ratio rises from 2.2$\times$ to 6.7$\times$, and the driver variance share rises from 83% to 98%. Driver mean norm grew modestly (1.21 $\rightarrow$ 1.46) rather than staying flat, consistent with better driver gradient SNR pushing harder against weight decay. Best val NLL also improves substantially (2.17 $\rightarrow$ 1.85), showing the N=512 sweep is data-limited. Off-diagonal cosines on the single N=5120 run are higher than the N=512 average (mean $|\cos|$ = 0.19 vs. 0.05); with only one seed at large N we cannot separate a real data-regime effect from seed-level variation and we do not attribute it. (ii) An L_1 penalty on the rows of W (or any group/row-norm sparsity regulariser) would drive distractor norms to exact zero without affecting drivers materially, at the cost of an extra hyperparameter and a deviation from the bare `nn.Linear(C,` $d_{\texttt{model}}$`)` we are auditing. (iii) Larger weight decay shrinks all norms, raising the driver/distractor ratio but at the cost of squeezing the drivers too. We do not employ either of these in the headline because the qualitative result—"the model down-weights distractors and sharpens that down-weighting with more data"—is already supported by the norm gap, the variance-share split, and the N-scaling probe.

3.7 Channel identifiability via linear probing

Following the procedure in Section 2.5, all per-channel-W_k encoders achieve $R^2 \approx 1.0$ at the encoder output (layer 0) and $R^2 \geq 0.84$ at the output of the third transformer block. SUM collapses to $R^2 \approx 0.24$ at every depth—channel identity is unrecoverable from the embedding it produces.

3.8 Channel masking

Zeroing one channel at test time (Table 9), the top-tier encoders show sharp, interpretable per-channel accuracy drops that match the outcome formula. SUM is nearly flat.

3.9 Convergence and wall-clock cost

A natural worry is that more involved encoders—an MLP stem, a projected positional layer, an orthogonality regulariser—might cost extra training time even when they don't change the ceiling. From the 300-epoch traces (val NLL recorded every 20 epochs, C=4, 5 seeds) we extract two

Table 8: Linear-probe channel recovery at C=4, seed 0, 300 epochs. Closed-form ridge probe $\mathbf{h} \mapsto \hat{x}$ from a frozen hidden state to the raw input $x \in \mathbb{R}^4$; held-out per-channel R^2. Layer 0 is the input embedding; layer 3 is the final transformer block. All per-channel-W_k encoders achieve near-perfect input recovery and preserve mean $R^2 \geq 0.84$ through three attention layers; SUM collapses to $R^2 \approx 0.24$ at every depth.

Encoder	Layer	R^2_{ch0}	R^2_{ch1}	R^2_{ch2}	R^2_{ch3}	mean
SUM	0 (input)	0.255	0.250	0.242	0.214	0.240
SUM	3 (final)	0.248	0.242	0.235	0.206	0.233
LINEAR	0 (input)	1.000	1.000	1.000	1.000	1.000
LINEAR	3 (final)	0.957	0.861	0.911	0.879	0.902
SUM-ORTHO	0 (input)	1.000	1.000	1.000	1.000	1.000
SUM-ORTHO	3 (final)	0.959	0.856	0.915	0.885	0.904
MLP	0 (input)	1.000	1.000	1.000	1.000	1.000
MLP	3 (final)	0.934	0.879	0.913	0.646	0.843
LINEAR-PPE	0 (input)	1.000	1.000	1.000	1.000	1.000
LINEAR-PPE	3 (final)	0.930	0.861	0.874	0.847	0.878
CONCAT	0 (input)	0.998	1.000	1.000	1.000	0.999
CONCAT	3 (final)	0.955	0.852	0.906	0.845	0.890

diagnostics: the epoch at which each run first reaches within 0.05 NLL of its own best, and the wall-clock seconds per epoch, normalised to LINEAR.

The five top-tier encoders reach within 0.05 NLL of their final value at the same trace point ($\sim$150 epochs) and cost essentially the same per-epoch wall time—the transformer backbone dominates, so encoder choice within this family is free. In particular, the MLP stem's 9× extra parameters do not slow convergence.

CI and CAT pay substantial wall-clock costs from their architectures, not their input encoders: $\sim$2.3× LINEAR at C=4 for CI (scaling to $\sim$8× at C=16) and $\sim$5× at C=4 for CAT (scaling with C^2T^2 to $\sim$17× at C=8). On the synthetic benchmark CAT also converges to its worse ceiling later than the top tier reaches its ceiling.

3.10 Positional projection: a small but real effect across C

LINEAR-PPE achieves the lowest val NLL at every C we test. The paired-seed design (Section 2.3) gives one Δ_s per seed; at 20 seeds the picture is:

- C=4: Δ = 0.041, paired t=6.60, p=2.6×10^{-6}, 95% bootstrap CI [+0.029, +0.053], 19/20 favour ppe.
- C=8: Δ = 0.026, paired p=3.7×10^{-4}, 95% bootstrap CI [+0.014, +0.038], 16/20.
- C=16: Δ = 0.016, paired p=0.036, 95% bootstrap CI [+0.002, +0.028], 14/20.

The gap shrinks with C but does not vanish; even at C=16, where the earlier 5-seed analysis read as null, the 20-seed paired test puts Δ above zero, just barely. The advantage is small in magnitude—$\sim$2% of baseline NLL at C=4, $\sim$0.7% at C=16—and the practical-significance reading is more subdued than the statistical-significance one would suggest in isolation.

Table 9: Test-time channel ablation at C=4, seed 0, 300 epochs. We zero a single channel's input at inference; rows show the resulting val accuracy. Δ is accuracy drop from the unmasked baseline. The outcome formula uses ch0 in two interaction terms, ch1 in one, and ch2/ch3 once each with smaller marginal effect; the per-channel-W_k encoders reproduce this ordering cleanly, whereas SUM, CI, and CAT are flatter.

encoder	no mask	mask ch0	mask ch1	mask ch2	mask ch3
SUM	0.093	0.082	0.089	0.084	0.094
Δ acc	—	−0.011	−0.004	−0.009	+0.001
CI	0.144	0.082	0.108	0.086	0.134
Δ acc	—	−0.062	−0.036	−0.058	−0.010
CAT	0.217	0.095	0.110	0.110	0.200
Δ acc	—	−0.122	−0.107	−0.107	−0.017
CONCAT	0.241	0.096	0.113	0.103	0.202
Δ acc	—	−0.145	−0.128	−0.138	−0.039
LINEAR	0.238	0.096	0.116	0.106	0.204
Δ acc	—	−0.142	−0.122	−0.132	−0.034
SUM-ORTHO	0.239	0.096	0.117	0.104	0.205
Δ acc	—	−0.143	−0.122	−0.135	−0.034
MLP	0.240	0.098	0.111	0.106	0.209
Δ acc	—	−0.142	−0.129	−0.134	−0.031
LINEAR-PPE	0.256	0.096	0.115	0.106	0.216
Δ acc	—	−0.160	−0.141	−0.150	−0.040

Direct measurement: orthogonalisation, not compression. We extract the per-channel projection matrix $W \in \mathbb{R}^{C\times d_{\text{model}}}$ and the effective positional basis $P \in \mathbb{R}^{T\times d_{\text{model}}}$ from trained LINEAR and LINEAR-PPE models (C=4, 20 seeds each; the `pospro_geometry` stage of our reproducer). For LINEAR, $P_{t,:} = \mathbf{p}(t)$ is the fixed sinusoidal basis; for LINEAR-PPE, $P_{t,:} = W_{\text{pos}}\mathbf{p}(t) + \mathbf{b}_{\text{pos}}$ is the learned-rotated basis. Two metrics suffice to discriminate the mechanism:

The two metrics give opposite verdicts on the two readings. The *orthogonalisation* reading is empirically confirmed: the fraction of P's energy that lies inside span(W) drops by $\sim$6.3× under LINEAR-PPE (3.4% → 0.5%; paired-difference 95% bootstrap CI on the reduction [−0.034, −0.024]), so the learned W_{pos} rotation actively pushes the positional encoding out of the channel subspace. The *compression* reading is contradicted: P's effective rank is *higher* under LINEAR-PPE (9.55 vs. 7.59), not lower—the rotation spreads positional energy across more directions, not fewer. The mechanism is rotation, not compression.

The redistribution is visible at the level of individual singular values: LINEAR's P has one dominant direction ($\sigma_0 \approx 52$) plus a long tail; LINEAR-PPE's is more evenly distributed ($\sigma_0 \approx 25$, with the next several around 11–13). Rotating the dominant positional mode out of span(W) inevitably redistributes its energy across other directions, which is the same effect read either as "orthogonalising position from channels" or as "flattening the singular spectrum of P".

We also computed a third candidate diagnostic—principal angles between span(W) and span(P)—but found it degenerate at this dimensionality: span(W) is only 4-dimensional, span(P) has rank $\sim$22 in a 64-dimensional ambient space, and span(W) is therefore trivially contained in span(P) for either encoder (all principal angles come out $\sim$0). Principal angles measure subset membership, not energetic overlap; the fraction-of-energy metric is the right tool here.

Table 10: Convergence speed and wall-clock cost (C=4, 300 epochs, 5 seeds, single GPU). Epochs-to-target is rounded to the trace resolution of 20 epochs. Cost is per-epoch wall time normalised to LINEAR.

encoder	best NLL	epoch to NLL +0.05	relative cost
LINEAR	2.170	∼152	1.00×
SUM-ORTHO	2.170	∼156	1.04×
MLP	2.171	∼144	1.01×
CONCAT	2.183	∼152	1.00×
LINEAR-PPE	2.116	∼148	0.99×
SUM	3.252	∼20 (plateau)	1.00×
CI	3.054	∼20 (plateau)	2.3×
CAT	2.360	∼180	5.2×

Table 11: Geometric measurement of the positional basis P and its overlap with the channel subspace span(W) (C=4, 300 epochs, mean over 20 seeds with 95% bootstrap CIs in brackets where seed variance is non-zero).

	LINEAR	LINEAR-PPE
Effective rank of P (entropy of $\tilde{\sigma}^2$)	7.59	9.55 $[9.17, 9.97]$
Fraction of $\|P\|_F^2$ within span(W)	0.034 $[0.029, 0.039]$	0.005 $[0.005, 0.006]$

With orthogonalisation confirmed, the C-dependent shrinkage of the gap admits a natural mechanistic reading: W_{pos} must rotate P out of a C-dimensional channel span using a fixed d^2_{model} parameter budget, and the rotation becomes increasingly over-constrained as C grows. At C=4 ($d_{\text{model}}-C = 60$ "free" dimensions for P to occupy) the rotation clearly helps; at C=16 ($d_{\text{model}}-C = 48$, plus more channel pressure on the residual stream overall) it helps less. We cannot fully separate this from two other plausible factors—the synthetic task has only four drivers regardless of C, so representational effort shifts elsewhere at large C; and seed-level optimisation variance is larger at higher C—but the over-constraint reading is the only one that directly invokes the empirically-supported mechanism.

d_{model} sweep. The model-width sweep (Section 3.3, Table 4) extends the LINEAR-PPE test to C=4 across $d_{\text{model}} \in \{64, 128, 256\}$. LINEAR-PPE achieves the lowest val NLL at every width tested. The paired gap to LINEAR is 0.050 at d=64 (p=0.012, 5/5), 0.011 at d=128 (p=0.56, 4/5; pulled down by a single anomalous seed), and 0.075 at d=256 (p=0.002, 5/5). Across d_{model}, 14 of 15 paired-seed comparisons favour LINEAR-PPE. The non-monotonic magnitude across d_{model} is not what either naive mechanism would predict, but it is at least clear that the rotation-based advantage persists across a 4× range of widths, including in the regime where the model is otherwise overfitting at the larger widths.

3.11 Real-data validation: ETTh1

Table 12 mostly confirms the synthetic ordering on ETTh1: the per-channel-W_k family (LINEAR, SUM-ORTHO, CONCAT, LINEAR-PPE, MLP) cluster within seed std of each other, CI underperforms, and SUM fails catastrophically (NLL near $\ln 32 \approx 3.47$).

Table 12: ETTh1 next-step bin prediction. 7 variates, T=160, K=32 quantile bins on the target OT (oil temperature) computed from the train split. Target is the next-step bin of OT; all 7 variates are inputs. d_{model}=56, 7 heads, 3 layers, d_{ff}=224; 300 epochs, cosine decay, mean ± std over 20 seeds (5 canonical + 15 paired-seed extras).

encoder	val NLL ↓	val acc ↑
SUM	3.668 ± 0.063	0.016 ± 0.020
CI	0.865 ± 0.038	0.664 ± 0.018
CAT	0.551 ± 0.019	0.785 ± 0.010
LINEAR	0.561 ± 0.017	0.786 ± 0.008
SUM-ORTHO	0.561 ± 0.017	0.784 ± 0.009
CONCAT	0.571 ± 0.019	0.783 ± 0.013
LINEAR-PPE	0.573 ± 0.014	0.776 ± 0.009
MLP	0.585 ± 0.020	0.788 ± 0.011

One difference from the synthetic ranking: CAT, which sits in the middle of the synthetic table, here posts the lowest mean NLL (0.551). Paired-difference tests at 20 seeds give a nuanced picture:

- Decisive: CAT beats MLP (paired p=1.4×10^{-7}, 20/20) and LINEAR-PPE (p=3×10^{-4}, 17/20).
- Marginal: CAT beats CONCAT (p=0.002, 16/20).
- Indistinguishable: CAT vs. LINEAR ($\Delta = 0.010$, paired p=0.14, 13/20) and CAT vs. SUM-ORTHO ($\Delta = 0.010$, paired p=0.10, 13/20).

Best-bin accuracies across the top tier are indistinguishable (0.78–0.79 with seed std ∼0.01). So CAT posts the lowest mean NLL but is statistically tied with LINEAR and SUM-ORTHO at the seed-level we test. Read carefully, the finding is not that channel-as-token decisively wins on real data, but that its synthetic-benchmark disadvantage closes and a small directional lead emerges—one that does not survive a paired test against the closest competitors. A plausible reading is that ETTh1's seven variates are all genuinely informative measurements of the same physical system, so the fine-grained cross-variate attention CAT enables can pay; whether that pays *enough* to prefer it over the per-channel-W_k default at matched wall-clock budget is much less clear.

4 Discussion

linear is not new. Stacking per-channel projections W_k and summing is `nn.Linear(C,` $d_{\texttt{model}}$`)`—the most obvious default. Our contribution is not proposing it but auditing it: at matched parameter budget, it matches every alternative (block-partitioned, orthogonality-regularised, nonlinear, channel-independent, channel-as-token) on the synthetic benchmark, and ties at the top of the per-channel-W_k tier on ETTh1. Only the shared-projection SUM consistently loses.

What fails about the shared-scalar baseline: information loss at the encoder. The failure of SUM is not gradual or capacity-limited; it is algebraic and complete. With shared scalar projection W and constant per-channel embeddings e_k, the encoder simplifies to

$$\mathbf{h}(t) = W\sum_k v_k(t) + \underbrace{\sum_k e_k}_{\text{constant}} + \mathbf{p}(t),$$

i.e. at every time step the encoder depends on the inputs only through their pointwise sum $S(t) = \sum_k v_k(t)$. The map $(v_0(t), \ldots, v_{C-1}(t)) \mapsto S(t)$ is many-to-one with a $(C-1)$-dimensional null space: at every position t, $C-1$ channels' worth of degrees of freedom are destroyed before the transformer sees the data. The per-channel embeddings e_k are constants and do not carry input-dependent information; they merge into a single offset and cannot tag which channel contributed what.

This loss is irrecoverable downstream. Any function the transformer computes on $\mathbf{h}(t)$ is a function of $S(t)$, so by the data-processing inequality the model's prediction can contain no more information about $(v_0, \ldots, v_{C-1})$ than S alone does. The 15× d_{model} scaling in Table 4 moves SUM's NLL by less than 0.005 precisely because the extra parameters never get to see the lost information.

The probe makes the floor explicit. For independent unit-variance channels at C=4, the correlation between any single channel v_k and the channel sum S is $\rho^2 = 1/C = 0.25$; that is the theoretical ceiling for linearly recovering one channel from S. The layer-0 probe recovers SUM's channels at $R^2 \approx 0.24$ (Table 8), essentially touching the floor. The "ceiling" SUM hits is therefore not an optimisation artefact but the information-theoretic limit of what survives a shared scalar projection followed by summation. The bottleneck is not summation per se—LINEAR also sums—but the *shared* projection, which collapses the per-channel directions into one before the sum.

Orthogonality is spontaneous. LINEAR's W_k reach mean off-diagonal cosine ≈0.042 with no penalty; the regulariser tightens to ≈0.010 without changing NLL. The task loss provides the gradient pressure.

Importance weighting is automatic. At C=8, driver channels get ∼2.3× larger seed-averaged $\|W_k\|$ norms and 84% of the embedding variance; at C=16 the norm ratio is ∼2.2× and per-channel variance contributions are ∼4× larger for drivers than for distractors. The driver/distractor gap is robust across two channel counts; the fine-grained ordering within drivers is descriptive only.

The residual stream carries channel identity end-to-end. Linear probes at layer 3 recover each channel to $R^2 \geq 0.84$; LayerNorm rescales but does not linearly mix coordinates. The deeper reason this works is the pre-LN transformer's residual structure: each block applies $h \leftarrow h + \text{Attn}(\text{LN}(h))$ and $h \leftarrow h + \text{FFN}(\text{LN}(h))$, so an unmodified copy of the input embedding is carried forward through every block. The per-channel projections injected at the input therefore persist into deep hidden states via the residual stream itself, not because attention or FFN sublayers learn to preserve them. Whatever near-orthogonal channel directions LINEAR establishes at the embedding layer are still linearly recoverable several blocks deep because LayerNorm only rescales them and the additive residual keeps them in the basis. This makes the geometric argument in this paper self-consistent: per-channel-W_k encoders write the channels into a near-orthogonal subspace at layer 0, and the architecture preserves that subspace by construction.

When nonlinearity helps and when it hurts. MLP is statistically worse than LINEAR at C=4, d_{model}=64 by a small margin ($\Delta = 0.022$, paired p=2×10^{-4}, 18/20) despite its 9× more encoder parameters; widening the model at C=4 makes things worse still ($\Delta = 0.067$ at d=128, $\Delta = 0.085$ at d=256, both paired $p < 10^{-8}$). MLP's encoder parameters scale as d^2_{model}, which exacerbates the overfitting that the N=512 regime already produces at $d \geq 128$ (Section 3.3). At C=16 the picture flips: paired tests (Section 3.2) show MLP ahead of LINEAR ($\Delta = 0.036$, p=2×10^{-4}, 18/20), SUM-ORTHO ($\Delta = 0.036$, p=7×10^{-4}), and CONCAT ($\Delta = 0.108$, p=5×10^{-12}); against LINEAR-PPE the gap is small and borderline ($\Delta = 0.020$, paired p=0.053, 12/20). A plausible reading: with many channels (drivers plus more distractors) the nonlinearity buys the encoder a useful gating on the channel vector that a linear projection cannot match. At 10× training data (Table 3) MLP's lead persists (18/20 to 20/20 paired diffs favouring MLP, paired $p \in [2{\times}10^{-13}, 4{\times}10^{-4}]$) but shrinks $\sim$3× to $\Delta = 0.010$–0.023 NLL, indicating that the N=512 gap is partly amplified by the data-limited regime. The Conclusion's recommendation still favours LINEAR as the default—the gap is small, shrinks further with more data, and the architecture cost is real—but MLP is a reasonable second choice when the channel count is high.

Projected positional encoding: small but real across C. The paired-seed analysis in Section 3.10 establishes the LINEAR-PPE gap over LINEAR as small but statistically clear at every C tested: $\Delta = 0.041$ (p=2.6×10^{-6}) at C=4, $\Delta = 0.026$ (p=3.7×10^{-4}) at C=8, $\Delta = 0.016$ (p=0.036) at C=16. The gap shrinks with C but does not vanish. A direct geometric measurement (Table 11) settles the mechanism: the fraction of the positional basis P's energy that lies inside the channel subspace span(W) drops $\sim$6.3× under LINEAR-PPE (3.4% → 0.5%); the learned W_{pos} rotation actively pushes P out of span(W). The orthogonalisation reading is confirmed; the compression reading is contradicted by the same measurement (P's effective rank is *higher* under LINEAR-PPE, 9.55 vs. 7.59). The rotation is genuine, and the mechanism is geometrically principled. In magnitude the advantage is small (about 2% of baseline NLL at C=4, less at larger C); we read it as worth using when the extra d^2_{model} parameters are cheap relative to the backbone, and unsurprising to ignore otherwise.

The C-dependent collapse of the gap (from $\Delta = 0.054$ at C=4 to $\Delta = 0.008$ at C=16) then admits a natural reading: W_{pos} must rotate P out of a C-dimensional channel span using a fixed d^2_{model} parameter budget, and the rotation becomes increasingly over-constrained as C grows. We cannot fully separate this from other factors at large C (driver/distractor pressure shifting representational effort elsewhere; larger seed-level optimisation variance), but it is the mechanism-internal explanation.

Architectural baselines. CI and CAT give each channel nominally more capacity at substantial wall-clock cost—$\sim$8× LINEAR at C=16 for CI, $\sim$17× at C=8 for CAT. CI underperforms throughout: its bottleneck is the head combining C streams. CAT is more interesting. On the synthetic benchmark, where drivers and distractors are mixed, it sits well below the per-channel-W_k tier (2.35 vs. $\sim$2.15 NLL at C=4). On ETTh1, where all seven variates are informative measurements of one physical system, the gap closes and CAT posts the lowest mean NLL. Paired-difference tests at 20 seeds (Section 3.11) place CAT decisively above MLP (p=1.4×10^{-7}) and LINEAR-PPE (p=3×10^{-4}), marginally above CONCAT (p=0.002), and *statistically tied* with LINEAR (p=0.14) and SUM-ORTHO (p=0.10). Best-bin accuracies in the top tier are indistinguishable. So CAT's synthetic-benchmark disadvantage closes on ETTh1 but its claimed lead doesn't survive a paired test against the closest competitors; the practical reading is that CAT is competitive on ETTh1, not clearly best.

No convergence-time penalty for involved encoders. MLP, SUM-ORTHO, and LINEAR-PPE reach within 0.05 NLL of their final value at the same trace point as LINEAR ($\sim$150 epochs) and cost the same per-epoch wall time. Within the family that uses per-channel W_k on a shared backbone, encoder choice does not change either the convergence schedule or compute budget.

Reading for non-numerical inputs. The mechanism is geometric and encoding-agnostic: per-field embedding tables are the discrete analog of per-channel W_k, and bag-of-embeddings pooling is the discrete SUM. The LINEAR-PPE finding—that projecting positional signals into a learned subspace beats adding them—transfers most readily, applying to any discrete-sequence model where position competes with token semantics for the same coordinates. We do not test any of this; the variance-contribution diagnostic in particular does not translate cleanly to discrete inputs and would need a probe-based analog.

Practical near-equivalence within the top tier. The paired tests reported above are powered by 20-seed paired-difference designs and resolve several small gaps that the unpaired analysis at 5 seeds called borderline. It is worth separating that statistical power from a practical claim. The absolute differences in NLL among the per-channel-W_k family (LINEAR, SUM-ORTHO, CONCAT, MLP, LINEAR-PPE) at C=4 span 0.02 NLL on a baseline of 2.16; LINEAR-PPE's lead over LINEAR is 0.041 NLL, about 2% of the baseline; MLP's lead over the linear family at C=16 is 0.011–0.025 in the data-rich regime. These are real gaps under our experimental design, but they are small relative to the spread induced by typical task-design choices (channel count, sequence length, label binning, etc.). We would not expect them to translate cleanly to practically significant differences on new tasks with new data distributions. Within the top tier the encoder choice is, in any practical sense, close to a free parameter.

5 Limitations

- The synthetic benchmark deliberately makes channel identity informative. On tasks with exchangeable channels, SUM may not be penalised.
- Main sweep at d_{model}=64, $C \in \{4, 8, 16\}$; ETTh1 at d_{model}=56, C=7. The d_{model} sweep covers all top-tier encoders plus SUM at $d \in \{64, 128, 256\}, C$=4, but all encoders overfit at $d \geq 128$ under the main-sweep training data (Table 4); the reported best-NLL numbers there are effective-early-stopping minima rather than converged values.
- The main sweep at N_{series}=512 is data-limited. 10× training data improves all top-tier encoders by ∼0.4 NLL at C=16 (Table 3); a similar gap exists at C=4 (2.17 → 1.85, see Section 3.6). The relative ordering of encoders is robust to data scale, but absolute inter-encoder gaps shrink ∼5× at 10× data. Readers should treat the N=512 inter-encoder gaps as upper bounds on the data-rich-regime gaps.
- CAT is skipped at C=16 on the synthetic benchmark for compute reasons ($\mathcal{O}((CT)^2)$ attention).
- LINEAR-PPE was tested at $C \in \{4, 8, 16\}$ on the synthetic benchmark and at C=7 on ETTh1.
- Most configurations are at 20 seeds (5 canonical + 15 paired extras). Bias ablation, linear probing, and channel masking remain at the canonical seeds noted in their tables.

6 Related work

Channel-as-token and channel-independent TSF. iTransformer (5) treats each variate as a token; PatchTST (4) runs independent backbones per channel; Crossformer (6) does both. These preserve per-channel capacity at the cost of longer sequences or per-channel compute. Our encoders share one token per time step.

Linear input projections in TSF. Many multivariate transformers use a linear stem (2, 3). We isolate this bare projection and compare it explicitly against the shared-scalar alternative and architectural baselines.

Orthogonality regularisation. Bansal et al. (7) use soft-orthogonality penalties for feature decorrelation. We apply the same idea to per-channel input projections and find it unnecessary: the task loss produces the same geometry.

7 Conclusion

On tasks where channel identity is informative, a per-channel linear projection (`nn.Linear(C,` $d_{\texttt{model}}$`)`) matches every more elaborate encoder we tested up to small, statistically real but practically modest, differences. The task loss drives the channel projections to near-orthogonality, and learned norms are larger for outcome-relevant channels. A shared-scalar projection has a capacity-independent ceiling. Nonlinearity, block partitioning, orthogonality penalties, and channel-independent architectures all end up close to the linear baseline: paired-difference analysis at 20 seeds resolves several small inter-encoder gaps as statistically clear, but the gaps span a 0.02–0.04 NLL range on a 2.16 NLL baseline—small enough that we would not expect them to translate cleanly to practically significant differences on new tasks with new data distributions.

Three variants deserve a footnote rather than a full carve-out, with caveats:

- LINEAR-PPE consistently leads the linear family at every C we tested (paired p from 2.6×10^{-6} to 0.036, gap shrinking from 0.041 to 0.016 NLL). A direct geometric probe shows the mechanism is positional–channel orthogonalisation rather than positional subspace compression. The rotation has a clean geometric justification and adds $d_{\text{model}}^2 + d_{\text{model}}$ parameters with no observable wall-clock cost.
- MLP pulls ahead of the linear family at C=16 under paired analysis (paired $p \in [3\times10^{-4}, 5\times10^{-12}]$ vs. LINEAR, SUM-ORTHO, CONCAT; gap shrinks $\sim$3× at 10× training data but stays statistically clear). It is worse than LINEAR at C=4, and worse than the linear family at C=4, $d_{\text{model}} \geq 128$ where it overfits hardest.
- CAT posts the lowest mean NLL on ETTh1, separating cleanly from MLP and LINEAR-PPE under paired tests but *statistically tied* with LINEAR and SUM-ORTHO ($p > 0.1$). It costs $\sim$5× wall clock at C=4 and substantially more at higher C.

Practical recommendation. Use `nn.Linear(C,` $d_{\texttt{model}}$`)` as the channel encoding; optionally project the sinusoidal positional encoding through a learned linear layer (LINEAR-PPE) since it has a clean geometric justification and only $d_{\text{model}}^2 + d_{\text{model}}$ extra parameters at negligible wall-clock cost. Do not reach for more elaborate encoders unless there is a real and strong reason in the task at hand—e.g., genuinely high channel count where the MLP gating may help, or a setting where the $\sim$5× wall-clock cost of channel-as-token attention is justified by a known cross-variate structure. Within the per-channel-W_k family the encoder choice is, for any practical purpose, close to a free parameter.

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